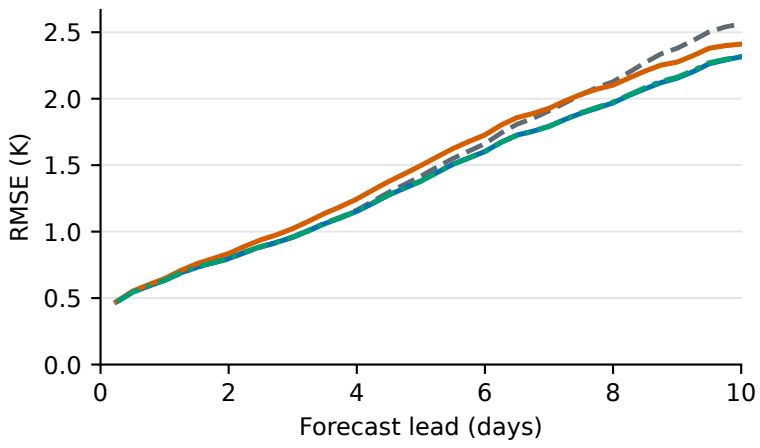
Forecast RMSE by variable

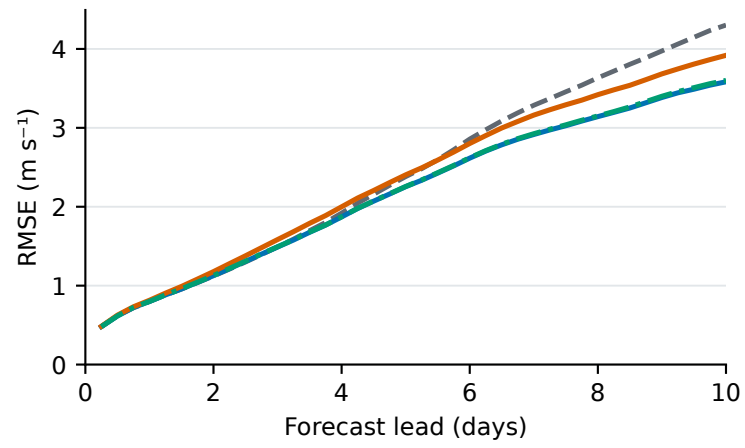
32 matched 2022 starts · 6-hourly leads through day 10 · residual models use SWA checkpoints

Original GC · MSE · AMSE λ=0.1 · Full AMSE

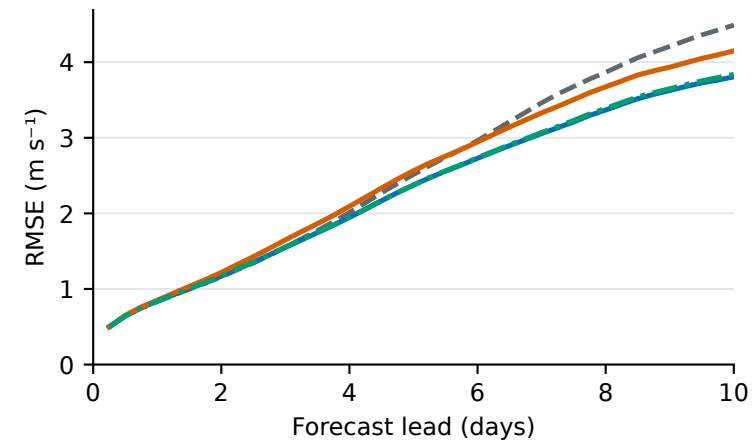
2 m temperature



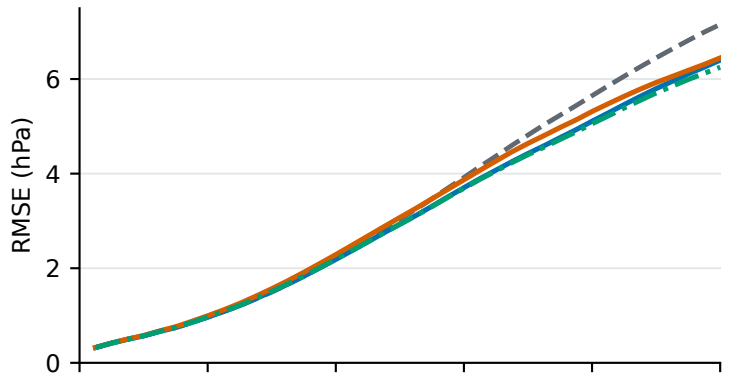
10 m zonal wind



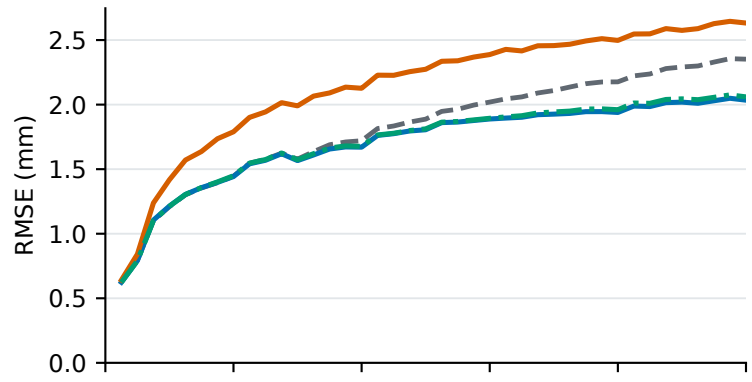
10 m meridional wind



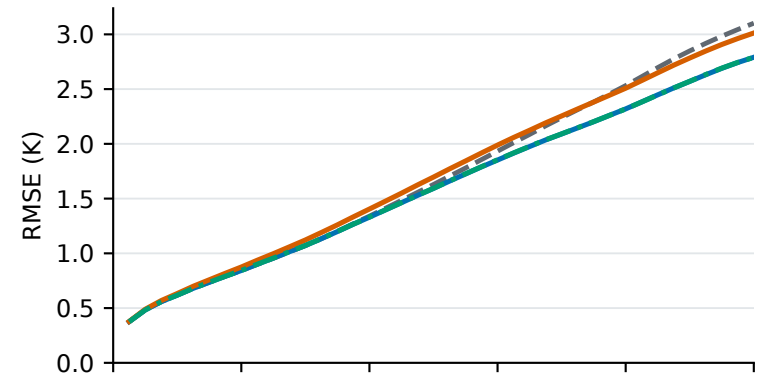
Mean sea-level pressure



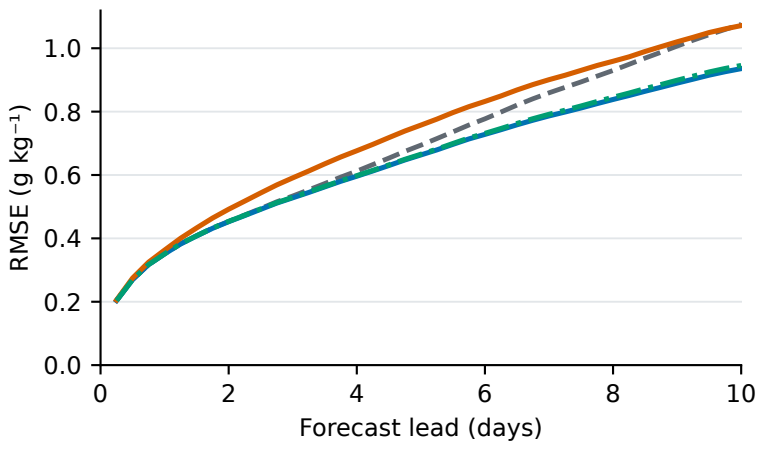
6 h precipitation



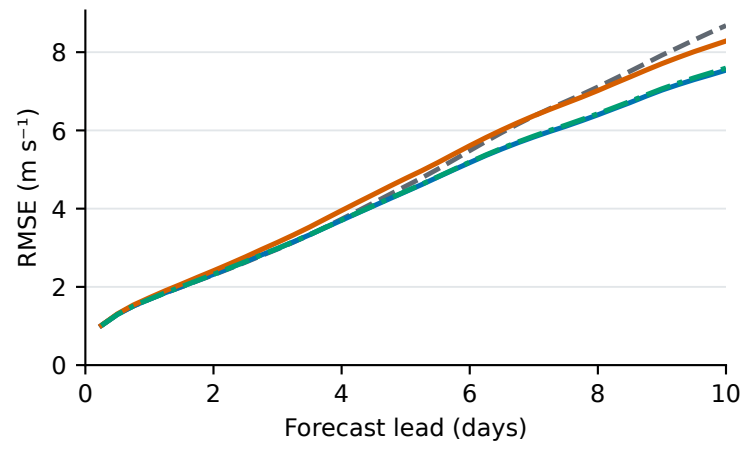
Temperature · all pressure levels



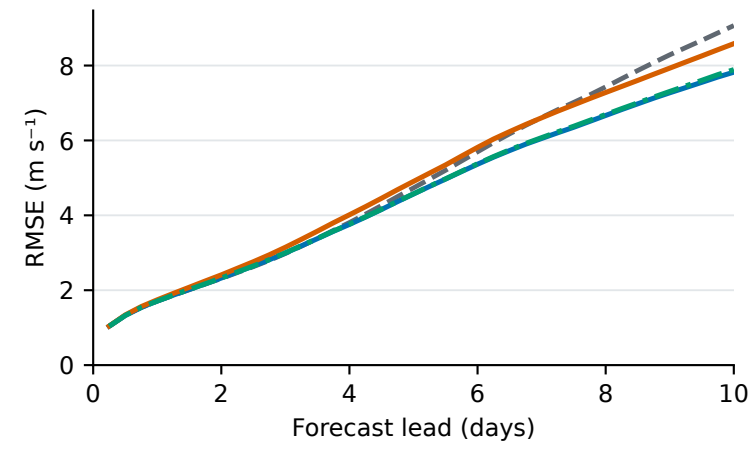
Specific humidity · all levels



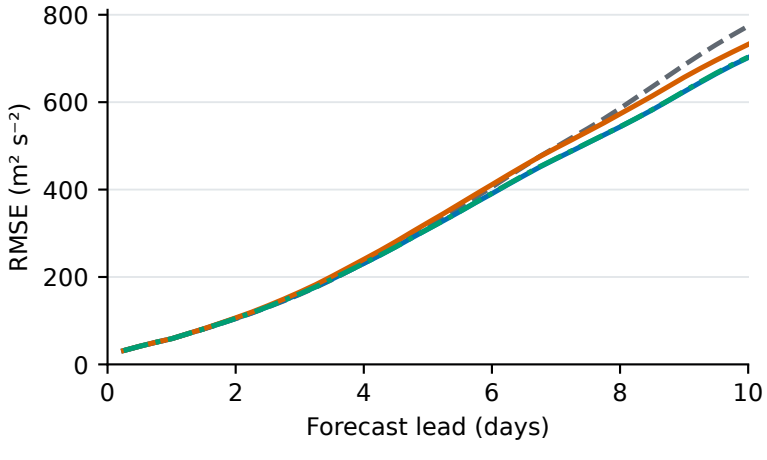
Zonal wind · all pressure levels



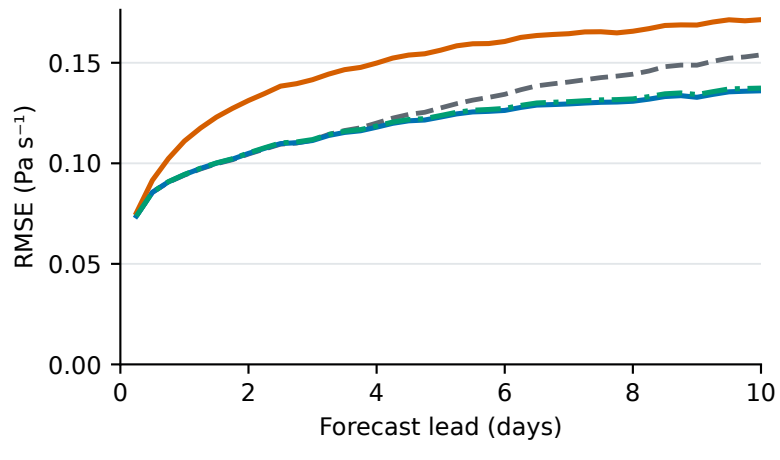
Meridional wind · all levels



Geopotential · all pressure levels



Vertical velocity · all levels



How to read these panels

Lower RMSE is better.
Each panel has its own physical units.

Upper-air groups pool squared error equally across all 13 pressure levels before taking the square root.

Global latitude-weighted errors pooled over the saved starts. These physical-variable RMSEs differ from the exact weighted GraphCast loss.

One common Original GC reference is shown; repeated baseline evaluations have small differences. No uncertainty intervals are available.